Case: Groupe Carrefour

Case Purpose

- Understand the reasons for Carrefour's performance trends and investors' skepticism
  - Showcase how to perform a step-by-step financial analysis
- Part I: Strategy analysis (Week 4)
- Part II: Financial analysis (Week 5)
  - Illustrate how to use Excel to conduct analysis
    - Preparation of condensed financial statements
    - The decomposition of ROE
    - The analysis of segment and non-financial information

Carrefour France

Carrefour China

Analyzing Carrefour's Strategy

- **10 mins** group discussion, please discuss the following questions:
  - What is Carrefour's competitive strategy?
  - What are Carrefour's core competencies?

Carrefour's Strategy

- Differentiation **and** cost leadership
  - Differentiation (hypermarket):
    - Broader assortment (food and non-food)
    - Wider choice of brands within one product category
    - Investing in customer loyalty (loyalty card, own brands and market campaign)
  - Cost leadership:
    - Assortment and price adjusted timely to meet changing customer preferences ("customer behavior database")
    - A well-developed logistics network
    - Economies of scale (bargaining power with suppliers)
    - Low-priced products under own brand

Carrefour's Strategy (cont'd)

- International growth*
  - Economies of scale in purchasing, logistics, and development of brand awareness (selling own branded products with same packaging worldwide)

*To achieve growth, large companies are often "forced" to expand overseas or enter other industries once they obtain a dominant position in domestic markets.

- Online growth
  - Competition from online retailers (but only 3% of total revenue in 2020, targeting 6% in 2022)
  - Continue to invest in online and omnichannel retailing

(in millions of euros)	2020	2019
France	34,135	34,765
Rest of Europe	21,239	20,999
Spain	9,058	8,799
Italy	4,172	4,516
Belgium	4,124	3,797
Poland	1,820	1,895
Romania	2,065	1,992
Latin America	13,245	14,665
Brazil	11,506	12,793
Argentina	1,739	1,872
Asia	2,100	1,968
Taiwan	2,100	1,968

From Carrefour 2020 Annual Report

Risks of Carrefour's Strategy

- **8 mins** group discussion, please discuss the following question:
  - What are the **risks** associated with Carrefour's competitive strategy?

Risks of Carrefour's Strategy

- Cost leader **and** differentiator
  - Mixed strategy
  - Hypermarkets are "stuck in the middle"

<u>Can you identify several points during the past two decades at which this happened?</u>

- During economic downturns in early 2000s
  - Focused too much on differentiation and improving margins rather than cutting prices – lost customers to price discounters

Risks of Carrefour's Strategy (cont'd)

- From 2012 to 2016 with CEO Plassat
  - Focus on cost-cutting and price competitiveness
  - Building Carrefour planet and spin-off discount retailer Dia

- From 2017 with CEO Bompard
  - Focus on cost cutting and global multi-store format
  - Online sales and omnichannel retailing

Risks of Carrefour's Strategy (cont'd)

- International growth
  - Different tastes and preferences of local customers
  - Risky economic environment of intercontinental hypermarkets, e.g., Latin America (customers price-sensitive, economic volatility, bureaucracy and strong government protectionism)
  - Severe price competition with other multinationals (i.e., Wal-Mart, Tesco) and strong local incumbents
  - During 2000s, international store portfolio too diversified.
    - Withdrew from 10 countries between 2011 and 2013.
    - Withdrew from China in 2019.

Ratio Analysis Pop Quiz

- What ratios can be used to measure a firm's profitability?

- What ratios can be used to measure a firm's efficiency in managing assets?

Ratio Analysis Pop Quiz

- What ratios can be used to measure a firm's profitability?
  - ROE, ROA, RNOA, net profit margin, net operating profit margin, gross profit margin, EBITDA margin, different expenses as percentage of revenue

- What ratios can be used to measure a firm's efficiency in managing assets?
  - Various asset turnover ratios, days in inventory, days in receivables, days in payables, cash conversion cycle

Questions:

2. Analyze Carrefour's performance for years 2016 to 2020 using financial statement data and segment data.

   A. What are the key drivers of Carrefour's performance changes since 2016?

   B. How does Carrefour's performance compare to that of Casino?

Financial Analysis of Carrefour

- Step 1: Prepare <u>condensed financial statements</u> using adjusted financial statements (Exhibit 3)
  - Allocate banking and financing income and expenses to investment profit: classify Carrefour's banking, customer credit, and real estate activities as non-operating investments (and associated income and expense items as investment profit).
  - Determine an appropriate tax rate: statutory tax rate (Exhibit 3)
  - Determine an appropriate operating cash percentage: assume 4%
  - Group equity=Ordinary shareholders' equity + Non-controlling interest in equity – Net assets held for sale

EXHIBIT 3 Carrefour's consolidated and adjusted income statements, balance sheets, and cash flow statements, 2015 to 2020 (in € millions)

The following financial statements are based on Carrefour's 2015–2020 annual reports and the author's calculations. The financials have been adjusted for off-balance operating leases and (partly) standardized.

Income statement (€ million)	2020	2019	2018	2017	2016	2015
Revenue – retail	70,719	72,397	76,000	78,897	76,645	76,945
Revenue – Carrefour banque	297	320	334	379	417	461
Financing fees and commissions	899	1,168	1,049	1,005	956	869
Total revenue	71,915	73,885	77,383	80,281	78,018	78,275
Cost of sales	–56,705	–58,049	–60,810	–62,692	–60,714	–60,838
SG&A expenses	–13,751	–14,302	–15,496	–16,431	–15,771	–15,675
Operating expenses – Carrefour banque	–260	–429	–291	–305	–325	–336
Net other operating income or expense	808	789	985	1,021	1,021	924
Non-operating investment income	162	248	255	249	208	248
Operating profit	2,169	2,142	2,026	2,123	2,437	2,598
Impairment losses	–295	–200	–202	–1,039	–130	–115
Restructuring costs	–93	–549	–727	–279	–154	–237
Other net non-recurring income or expense	–119	–302	–223	–78	–183	1
Interest income	6	8	16	28	32	36
Interest expense	–317	–377	–395	–501	–574	–566
Profit before taxes	1,351	722	495	254	1,428	1,717
Tax expense	–488	–490	–541	–622	–470	–557
Tax expense – Carrefour banque	–5	–7	1	2	–12	–20
Profit/loss to non-controlling interest	–190	–182	–216	–169	–148	–143
Profit/loss from continued operations	668	43	–261	–535	798	997

Financial Analysis of Carrefour (cont'd)

- Step 2: Calculate profitability ratios

Analysing Carrefour's Profitability

- **10 mins <u>group discussion</u>**, please discuss the following questions:
  - What are the primary drivers of the company's performance decline since 2016? *(HINT: ROE, RNOA and components)*
  - How does Carrefour's performance compare to the performance of Casino?

Carrefour's Profitability

Negative ROE/RNOA in 2017-2019 potentially driven by non-recurring/transitory items, e.g., impairment losses, restructuring costs.

	2020	2019	2018	2017	2016
Return on equity	7.5%	2.0%	-0.4%	-3.0%	8.4%
Return on net operating assets	1.5%	-1.7%	-3.3%	-4.4%	2.2%

Income statement (€ million)	**2020**	**2019**	**2018**	**2017**	**2016**	**2015**
Revenue – retail	70,719	72,397	76,000	78,897	76,645	76,945
Revenue – Carrefour banque	297	320	334	379	417	461
Financing fees and commissions	899	1,168	1,049	1,005	956	869
Total revenue	71,915	73,885	77,383	80,281	78,018	78,275
Cost of sales	–56,705	–58,049	–60,810	–62,692	–60,714	–60,838
SG&A expenses	–13,751	–14,302	–15,496	–16,431	–15,771	–15,675
Operating expenses – Carrefour banque	–260	–429	–291	–305	–325	–336
Net other operating income or expense	808	789	985	1,021	1,021	924
Non-operating investment income	162	248	255	249	208	248
Operating profit	2,169	2,142	2,026	2,123	2,437	2,598
Impairment losses	–295	–200	–202	–1,039	–130	–115
Restructuring costs	–93	–549	–727	–279	–154	–237
Other net non-recurring income or expense	–119	–302	–223	–78	–183	1
Interest income	6	8	16	28	32	36
Interest expense	–317	–377	–395	–501	–574	–566
Profit before taxes	1,351	722	495	254	1,428	1,717
Tax expense	–488	–490	–541	–622	–470	–557
Tax expense – Carrefour banque	–5	–7	1	2	–12	–20
Profit/loss to non-controlling interest	–190	–182	–216	–169	–148	–143
Profit/loss from continued operations	668	43	–261	–535	798	997

Carrefour's Profitability (cont'd)

Ratio	2020	2019	2018	2017	2016
ROE (after excluding non-recurring items after tax)	10.5%	8.8%	6.8%	6.0%	9.8%
RNOA (after excluding non-recurring items after tax)	3.2%	2.1%	0.7%	0.7%	3.0%

- Some may argue certain items are recurring (e.g., restructuring costs, impairment losses).
- In remaining analyses, we **assume all the these are non-recurring**, thus excluded from profit calculation.

Decompose ROE

	2020	2019	2018	2017	2016
Net profit margin (ROS)	1.7%	1.4%	1.0%	0.9%	1.5%
× Asset turnover	1.44	1.43	1.49	1.53	1.53
= Return on assets (ROA)	**2.4%**	**2.0%**	**1.6%**	**1.4%**	**2.2%**
× Financial leverage	4.30	4.42	4.35	4.28	4.43
= Return on equity (ROE)	**10.5%**	**8.8%**	**6.8%**	**6.0%**	**9.8%**

- Its ROE and ROA dropped in 2017, but gradually recovered since then.
  - The drop in 2017 was due to a drop in net profit margin, which gradually recovered.
  - Its asset turnover decreased slightly since 2017.
  - Its financial leverage remained stable.

Decompose RNOA

Ratio	2020	2019	2018	2017	2016
Net operating profit margin	0.9%	0.6%	0.2%	0.2%	0.8%
× Net operating asset turnover	3.42	3.47	3.63	3.68	3.66
= Return on net operating assets	**3.2%**	**2.1%**	**0.7%**	**0.7%**	**3.0%**

- Similarly, RNOA dropped in 2017 and 2018, but gradually recovered since then.
  - The drop in RNOA was due to a drop in NOPAT margin, which gradually recovered – consistent with cost savings.
  - Its net operating asset turnover declined since 2017 – concerning and suggesting unsuccessful in price competitiveness.

Carrefour vs. Casino

	Carrefour				Casino			
Year	**2020**	**2019**	**2018**	**2017**	**2020**	**2019**	**2018**	**2017**
NOPAT margin	0.9%	0.6%	0.2%	0.2%	2.7%	1.9%	2.1%	2.5%
× Net operating asset turnover	3.46	3.45	3.67	3.74	2.18	2.09	2.26	2.07
= RNOA	3.2%	2.1%	0.7%	0.7%	5.9%	4.0%	4.7%	5.2%

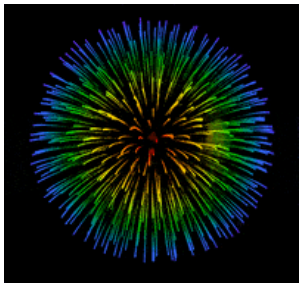
- Carrefour has much lower NOPAT margin and thus lower RNOA than Casino, although its operating asset turnover was higher.

- *Question: Given Carrefour's operating asset turnover (3.5) in 2020, at what NOPAT margin would Carrefour earn a similar RNOA to its peer Casino (6%)? Is this feasible?*

1.7% NOPAT margin to achieve 6% of RNOA, or a 0.8% increase. Given the 2020 revenue level (€70 billion), it is equivalent to an increase of €560 million (after tax) operating profit.
Hence, Carrefour's planned additional cost savings of €2.4 billion could be sufficient to help it return to a level of profitability comparable to its peers.

Analysing Carrefour's NOPAT Margin

- **10 mins** group discussion, please discuss the following question:
  - What are the primary drivers of Carrefour's low NOPAT margin? *(Hint: Which expenses were high?)*
  - How did operating profit margins vary across geographic regions and develop over time (within regions)?



Thank you for taking this course!

Your feedback makes a real difference:

- You said to give you pop quiz. **I DID!**
- You said to cover ESG topics. **I DID!**
- You said to choose your own industry for group project. **I DID!**

Don't miss out on your chance to let us know what you think.
To take the survey now, please scan the QR code:

Tell me what rocked and what flopped — I promise I won't cry (much)!

or go directly to https://lse.surveys.evasysplus.co.uk

Components of NOPAT Margin

	2020	2019	2018	2017	2016
Line items as a percent of sales					
Sales	100.0%	100.0%	100.0%	100.0%	100.0%
Interest expense	-0.4%	-0.5%	-0.5%	-0.6%	-0.7%
Tax expense	-0.7%	-0.6%	-0.6%	-0.6%	-0.8%
Operating expense line items as a percent of sales (by function)					
Cost of sales	-80.2%	-80.2%	-80.0%	-79.5%	-79.2%
Selling, general, and admin. expense	-19.4%	-19.8%	-20.4%	-20.8%	-20.6%
Operating expense line items as a percent of sales (by nature)					
Personnel expense	-10.3%	-10.4%	-10.7%	-10.9%	-10.8%
Depreciation and amortization	-2.9%	-2.9%	-3.4%	-3.4%	-3.2%

- Increasing cost of sales between 2016 and 2020 by 1% -- quite substantial given its thin NOPAT margin of 0.9% in 2020. Likely due to intense competition (lower price).
- Successful in reducing SG&A costs between 2018 and 2020, in particular, personnel expenses (lower than Casino's 11-12%).
- In 2019 and 2020, lower D&A also contributed to improved margin.

Segment Operating Profit Margin

Operating profit margin (Recurring operating income/Net revenue)	France	Rest of Europe	Latin America	Asia
2020	1.8%	3.3%	5.9%	4.5%
2019	1.6%	3.1%	5.7%	4.2%
2018	1.3%	3.2%	5.6%	0.8%
2017	1.9%	3.2%	4.5%	0.1%
2016	2.9%	3.5%	4.9%	-0.9%

- In **Asia**, average operating profit margins approached zero in 2016–2018, but improved after Carrefour pulled out of China (in 2019).
- During 2018–2020, Carrefour's operating performance suffered most in its home market **France** – intense competition. Margins improved in 2019 and 2020 thanks to cost savings.
- Trends in segment margins are consistent with the idea that disposing of underperforming Chinese assets and cost savings in the home market contributed to Carrefour's recent NOPAT margin improvements.

Analysing Carrefour's Asset Turnover

- **10 mins** group discussion, please discuss the following questions:
  - What are the primary driver(s) of Carrefour's operating asset turnover between 2016 and 2020?

  (Hint: Working capital turnover and non-current operating assets turnver)

Artificially increased due to a large impairment in 2017.

Improvement in inventory turnover.

Components of Operating Asset Turnover

Year	2020	2019	2018	2017	2016
Operating working capital/Sales	-6.0%	-6.3%	-6.3%	-6.3%	-6.4%
Net non-current tangible and intangible assets/Sales	35.0%	34.9%	33.6%	33.2%	33.4%
PP&E/Sales	21.8%	21.8%	21.3%	21.0%	20.8%
Operating working capital turnover	-16.60	-15.79	-15.79	-15.92	-15.59
Net non-current asset turnover	2.85	2.87	2.97	3.01	2.99
PP&E turnover	4.60	4.58	4.70	4.76	4.82
Trade receivables turnover	27.23	27.81	28.75	29.05	30.96
Days' receivables	13.4	13.1	12.7	12.6	11.8
Inventories turnover	10.13	9.67	9.48	9.13	9.06
Days' inventories	36.0	37.7	38.5	40.0	40.3
Trade payables turnover	4.33	4.18	4.16	4.11	4.18
Days' payables	84.3	87.4	87.8	88.7	87.3
Cash conversion cycle	-34.9	-36.6	-36.6	-36.2	-35.2

A reduction on Carrefour's reliance on supplier finance, negative effect on operating asset turnover.

Decline in non-current asset turnover.

Question

- Which geographic segments contributed most to the increase in operating asset turnover?
- Is there a measure for efficiency of utilizing non-current assets that is <u>not affected</u> by non-operating factors (e.g., impairments, or price of real estates) ?

Note: Store productivity=Revenue/Store area size.

Segment Asset Turnover

	France	Rest of Europe	Latin America	Asia
Total asset turnover				
2020	1.69	1.84	1.69	1.80
2019	1.73	1.93	1.88	1.24
2018	1.81	2.07	1.94	2.67
2017	1.79	1.99	2.18	2.59
2016	1.83	1.91	2.17	2.58
Revenue per square metre				
2020	6.22	3.61	4.97	2.01
2019	6.31	3.75	5.72	1.06
2018	6.30	3.76	5.62	2.04
2017	6.24	3.82	6.76	2.15
2016	6.30	3.50	6.32	2.25

- Store productivity is highest in France and lowest in Asia.
- In 2018-2020, Store productivity decreased in Latin America (but operating profit margin improved): Carrefour transformed Latin American hypermarkets into cash-and-carry stores. The ratios suggest that the new cash-and-carry stores have higher margins but lower turnover than the hypermarkets.

Geographic Segment (Exhibit 2)

	Fiscal year	France	Rest of Europe	Latin America	Asia
Number of hypermarkets	2020	248	456	185	172
	2019	248	455	188	175
	2018	247	452	189	372
	2017	247	460	193	365
	2016	243	439	316	374
	2015	242	489	304	369
Number of supermarkets	2020	1,179	1,873	151	10
	2019	1,071	1,798	150	77
	2018	1,056	1,776	147	73
	2017	1,060	1,756	147	58
	2016	1,062	1,777	168	38
	2015	1,003	2,096	168	29
Number of convenience stores	2020	4,018	3,156	530	66
	2019	3,959	2,646	530	0
	2018	3,918	2,511	516	29
	2017	4,267	2,446	521	41
	2016	4,222	2,312	468	27
	2015	4,263	2,464	404	8
Number of cash & carry stores	2020	147	68	215	0
	2019	146	60	193	0
	2018	144	49	173	0
	2017	144	42	153	2
	2016	143	13	0	2
	2015	142	18	0	0

Financial Analysis – Conclusion

- *Competition.* The intense competition in Carrefour's home market harms its NOPAT margins.
- *Cost control.* Carrefour succeeded in achieving cost savings that helped improve operating margins in 2019 and 2020. However, although Carrefour announced that it had succeeded in realizing €3 billion of cost savings, which should increase after-tax profit by €2 billion, or 2.9% of revenue. However, NOPAT margins improved by a much lower percentage – revenue decreases partly offset the impact of cost savings. More is needed to help NOPAT margins reach a more acceptable level of 2%.
- *Geography.* Carrefour's Asian segment (primarily China) has strongly underperformed, justifying the disposal of Chinese activities in 2019. Carrefour's withdrawal from China contributed to its performance improvement.

Financial Analysis– Conclusion (cont'd)

What other actions can Carrefour take to improve its margins?

- Further increasing its focus on cash-and-carry and convenience stores, reducing the investments in hypermarkets.
- Finding more cost-efficient ways to improve online presence (such as entering into strategic deals with existing online retailers).
- Moving to a "franchise" model for underperforming French hypermarkets, giving local managers more ownership and freedom to address local challenges.

Thank you for attending AC444. See you at the Group Presentation!